

Three-Configuration Evaluation Report

1. Executive Summary

This report compares the best former hybrid configuration, the best pure-LLM configuration, and the new two-stage direction-first hybrid.

The evaluation contains:

- 57 transition pairs
- 159 evaluated vital-sign observations
- 58 ground-truth decreases
- 56 ground-truth increases
- 45 ground-truth stable observations

The primary metric is vital-level direction accuracy.

Configuration	Correct directions	Vital-level accuracy	Normalized RMSE
Former hybrid: GPT-5, 3-shot static	102/159	64.15%	3.520
Pure LLM: GPT-5.5, 3-shot kind-hint matched	113/159	71.07%	3.474
Two-stage hybrid: GPT-5, 3-shot static	106/159	66.67%	3.373

The pure LLM remains the best overall configuration. It has the highest direction accuracy and the best normalized MAE while maintaining substantially better balance across increase, decrease, and stable categories.

The two-stage hybrid improves over the former hybrid by 4 correct observations, or 2.52 percentage points. It is also the best configuration for `no_action` and for recognizing true increases and decreases. However, it **overcorrects** the former hybrid's conservative behavior and frequently predicts a change when the ground truth is stable.

2. Compared Configurations

2.1 Former Hybrid

- Model: GPT-5
- Prompting: 3-shot static
- Structure: rule-based prediction plus an LLM-supplied signed residual

2.2 Pure LLM

- Model: GPT-5.5
- Prompting: 3-shot kind-hint matched
- Structure: LLM directly predicts absolute post-transition vital values

2.3 Two-Stage Hybrid

- Model: GPT-5
- Prompting: 3-shot static with fixed, role-based demonstrations
- Stage 1: LLM predicts `increase`, `decrease`, or `stable`

- Stage 2: LLM predicts an unsigned magnitude after the direction is fixed
- Rule engine: used as a magnitude reference and fallback

3. Evaluation Method

Direction categories use the repository's vital-specific dead zones:

Vital	Stable dead zone
HR	+/-5 bpm
BP_sys	+/-5 mmHg
BP_dia	+/-5 mmHg
RR	+/-2 breaths/min
O2Sat	+/-2 percentage points
T	+/-0.2 degrees

4. Overall Direction Performance

4.1 Class Precision and Recall

Configuration	Class	Precision	Recall	Support	Predicted
Former hybrid	Decrease	84.38%	46.55%	58	32
Former hybrid	Increase	76.92%	71.43%	56	52
Former hybrid	Stable	46.67%	77.78%	45	75
Pure LLM	Decrease	84.21%	55.17%	58	38
Pure LLM	Increase	79.31%	82.14%	56	58
Pure LLM	Stable	55.56%	77.78%	45	63
Two-stage hybrid	Decrease	66.67%	75.86%	58	66
Two-stage hybrid	Increase	66.67%	85.71%	56	72
Two-stage hybrid	Stable	66.67%	31.11%	45	21

The former hybrid is conservative. It predicts stable 75 times even though stable appears only 45 times. This gives it high stable recall but low decrease recall.

The pure LLM is the most balanced configuration. Its predicted class distribution is closer to the ground-truth distribution, and it has the best increase precision while retaining strong stable recall.

The two-stage hybrid predicts non-stable directions much more often. This substantially improves increase and decrease recall, but stable recall falls to 31.11%.

5. Confusion Matrices

Rows are ground truth. Columns are engine predictions.

5.1 Former Hybrid

Ground truth	Predicted decrease	Predicted increase	Predicted stable	Support
Decrease	27	5	26	58
Increase	2	40	14	56
Stable	3	7	35	45

5.2 Pure LLM

Ground truth	Predicted decrease	Predicted increase	Predicted stable	Support
Decrease	32	6	20	58
Increase	2	46	8	56
Stable	4	6	35	45

5.3 Two-Stage Hybrid

Ground truth	Predicted decrease	Predicted increase	Predicted stable	Support
Decrease	44	8	6	58
Increase	7	48	1	56
Stable	15	16	14	45

6. Main Direction Failures

Error type	Former hybrid	Pure LLM	Two-stage hybrid
Decrease -> stable	26	20	6
Increase -> stable	14	8	1
Stable -> decrease	3	4	15
Stable -> increase	7	6	16
Decrease -> increase	5	6	8
Increase -> decrease	2	2	7
Total direction errors	57	46	53

6.1 Former Hybrid

The former hybrid's main problem is conservative stable prediction:

- 40 of 57 errors are non-stable observations predicted as stable.
- Decrease-to-stable alone accounts for 26 errors.
- Decrease recall is only 46.55%.

This supports the earlier conclusion that the rule-plus-residual architecture anchors the LLM too strongly to small or stable rule-based changes.

6.2 Pure LLM

The pure LLM still exhibits some stable bias:

- 20 decreases are predicted as stable.
- 8 increases are predicted as stable.

However, it makes only 10 stable-to-non-stable errors and therefore maintains the best overall balance. Its principal remaining direction failure is decrease-to-stable.

6.3 Two-Stage Hybrid

The two-stage design solves most of the former conservative failure:

- Decrease-to-stable falls from 26 to 6.
- Increase-to-stable falls from 14 to 1.

The new dominant error is the reverse:

- 31 of 53 errors are stable observations predicted as changing.
- 16 stable observations are predicted as increases.
- 15 stable observations are predicted as decreases.

The design therefore changes the bias from "too stable" to "too active." It also increases complete direction reversals: 15 increase/decrease reversals, compared with 7 for the former hybrid and 8 for the pure LLM.

7. Performance by Vital

Vital	Support	Former hybrid	Pure LLM	Two-stage hybrid
HR	33	42.42%	69.70%	75.76%
BP_sys	37	83.78%	83.78%	70.27%
BP_dia	35	60.00%	68.57%	62.86%
RR	22	54.55%	45.45%	45.45%
O2Sat	30	73.33%	76.67%	73.33%
T	2	100.00%	100.00%	50.00%

The two-stage design produces a large HR improvement but loses performance on BP_sys and RR.

8. Performance by Common Action Type

Action type	Support	Former hybrid	Pure LLM	Two-stage hybrid
no_action	42	47.62%	54.76%	59.52%
fluid_bolus	30	63.33%	86.67%	56.67%
oxygen_support	22	77.27%	81.82%	63.64%
airway_management	19	63.16%	57.89%	63.16%

The two-stage hybrid is the strongest configuration for `no_action`, improving by 11.90 percentage points over the former hybrid. This is evidence that an explicit direction decision helps when the rule engine would otherwise favor no change.

The pure LLM is substantially stronger for `fluid_bolus` and `oxygen_support`. The two-stage direction prompt appears to infer clinically plausible change too readily in cases whose evaluated ground truth remains within the stable dead zone.

9. Interpretation

Best overall configuration

The pure LLM is currently the strongest general-purpose configuration:

- Highest vital-level direction accuracy: 71.07%
- Strongest balance among all three direction categories

Value of the two-stage hybrid

The two-stage hybrid demonstrates that direction-first prediction can address the former hybrid's conservative bias:

- Best decrease recall
- Best increase recall
- Best `no_action` accuracy
- Lowest normalized RMSE
- Complete reasoning for both stages

However, its current direction stage is insufficiently calibrated for stable observations. Its lower overall accuracy